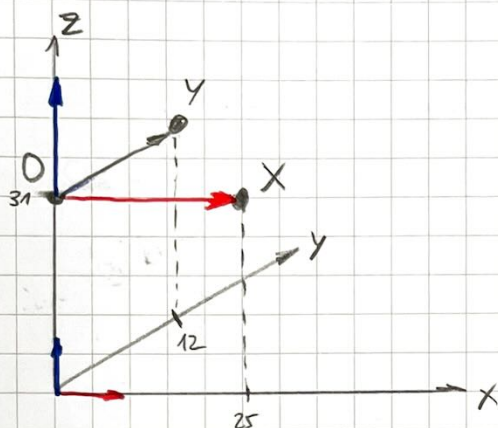


07.01.2025

# TE Robotique

Florent Jagvier

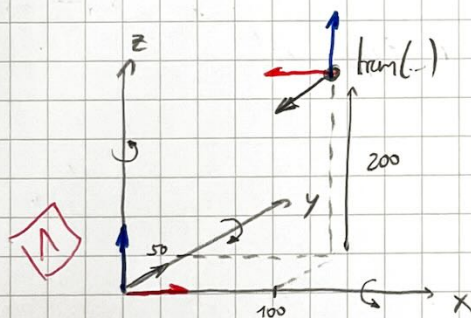
Q14



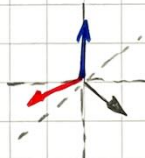
$$\text{frame} = (0, 0, 31, 0, 0, 0)$$

Q10 4.5

a)

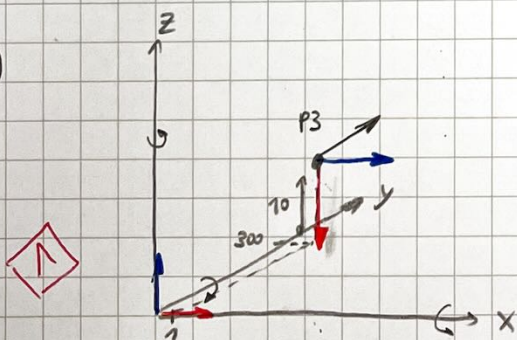


| $x$ | $y$ | $z$ | $rx$ | $ry$ | $rz$ |
|-----|-----|-----|------|------|------|
| 100 | 50  | 200 | 0    | 0    | 180  |
| 0   | 0   | 400 | 0    | 0    | -135 |

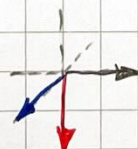


$$P_2 = (0, 0, 400, 0, 0, -135) \checkmark$$

b)



| x | y   | z    | rx | ry | rz  |
|---|-----|------|----|----|-----|
| 1 | 300 | 10   | 0  | 90 | 0   |
| 1 | 400 | -190 | 90 | 0  | -90 |

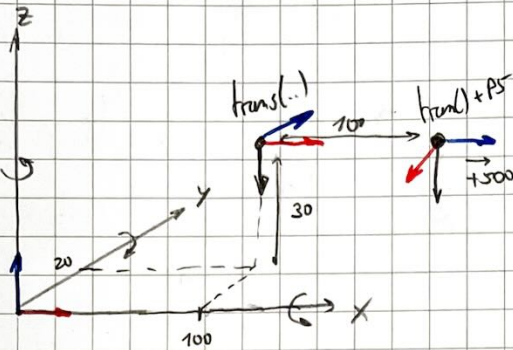


$$P_4 = (1, 400, -190, 90, 0, -90)$$



c)

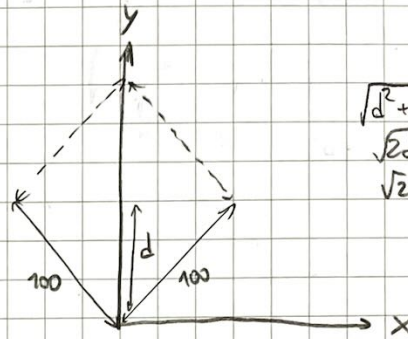
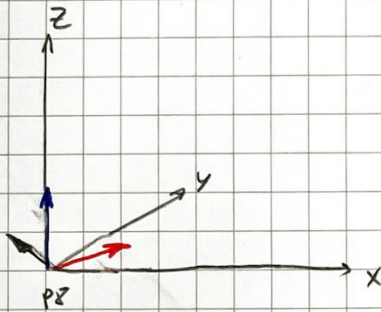
2



|            | x   | y  | z  | r <sub>x</sub> | r <sub>y</sub> | r <sub>z</sub> |
|------------|-----|----|----|----------------|----------------|----------------|
| trans(...) | 100 | 20 | 30 | -90            | 0              | 0              |
| +P5        | 200 | 20 | 30 | -90            | 90             | 0              |
| +P6        | 700 | 20 | 30 | -90            | 90             | 29             |

$$P7 = (700, 20, 30, -90, 90, 29) \quad \checkmark$$

d)



$$\begin{aligned} \sqrt{d^2 + d^2} &= 100 \\ \sqrt{2d^2} &= 100 \\ \sqrt{2}d &= 100 \\ d &= \frac{100}{\sqrt{2}} \end{aligned}$$

$$x = 0 \quad \checkmark$$

$$y = 2 \cdot d = \frac{200}{\sqrt{2}} \quad 2 \cdot \frac{\sqrt{2}}{2} \cdot 100 = 100 \cdot \sqrt{2}$$

$$P9 = (0, \frac{200}{\sqrt{2}}, 100, 0, 0, 45)$$